

Express-Session Detailed Guide

1. What is express-session?

express-session is middleware for Express.js applications. It helps store session data on the server so your app can remember users between HTTP requests.

2. Why Sessions Are Needed

HTTP is stateless. That means each request is independent. Sessions allow login state, carts, user preferences, and temporary data to persist.

3. How It Works

1) User visits website. 2) Server creates a session object. 3) Server sends session ID cookie to browser. 4) Browser sends cookie on next requests. 5) Server finds matching session data.

4. Basic Setup

```
const express = require('express');
const session = require('express-session');
const app = express();

app.use(session({
  secret: 'my-secret-key',
  resave: false,
  saveUninitialized: false,
  cookie: { maxAge: 60000 }
}));

app.get('/', (req,res)=>{
  req.session.views = (req.session.views || 0) + 1;
  res.send('Views: ' + req.session.views);
});

app.listen(3000);
```

5. Important Options

secret: signs cookie.
resave: save even if unchanged.
saveUninitialized: save empty sessions.
cookie.maxAge: expiry time.
cookie.secure: HTTPS only.

6. Production Best Practice

Default MemoryStore is not recommended for production. Use Redis, MongoDB, or SQL-based session stores.

7. Common Use Cases

User login systems, admin dashboards, carts, multi-step forms, flash messages.

8. express-session vs JWT

Sessions store data on server. JWT stores data in token on client side. Sessions are simpler for many web apps.

9. Summary

express-session helps your Express app remember users securely across requests.